# HYBRID PACKET LOSS CONCEALMENT FOR REAL-TIME NETWORKED SPEECH APPLICATIONS

*Messana Silvia, Viviani Marco*

Dipartimento di Elettronica, Informazione e Bioingegneria (DEIB), Politecnico di Milano
Piazza Leonardo Da Vinci 32, 20122 Milano, Italy
[silvia.messana, marco1.viviani]@mail.polimi.it

## ABSTRACT

Audio packet loss concealment (PLC) consists in correcting the corrupted messages in audio stream communications over internet protocol (IP) caused by data transmission failures. Nowadays voice over IP telephony and teleconference systems are essential in both professional and personal life, nonetheless are subject to data losses and latency. Therefore we need effective PLC techniques.
In this report we present an hybrid technique where a feed-forward neural network and an autoregressive (AR) model cooperate in order to reconstruct the lost packets. The model under consideration is **PARCnet** (Parallel AutoRegressive Conv-net), a recent PLC technique which has demonstrated excellent results in the field of real-time music networks. The main goal of this work is to test the performance of such a model in the speech reconstruction domain. To evaluate the results we use two metrics: PLCMOS, a novel data-driven metric used to evaluate the performance of PLC systems and the word accuracy (WAcc) that exploits Microsoft Azure Cognitive Services.
As expected the model didn't perform as in the original framework, but results definitely enrich the field of PLC for future works.

## 1. INTRODUCTION

In the past years many different techniques of PLC have been proposed and led to a flourishing literature that will be examined in the following. Some of the classic techniques of PLC that set the ground for more appealing ones are based on waveform substitution and on modifying the coding paradigm.
The former [1] exploits pattern matching: the missing packet is substituted with the frame from the history that best matches the one preceding the loss. Alternatively, pitch preceding the loss is estimated; samples from the last period are repeated in a cyclic fashion, instead of the missing packet.
The latter can be done using various methods. For example in [2] the odd and even samples are transmitted in different packets: when a packet is missing audio can be restored via interpolation. Another example is [3], where the most significant bits of the pulse code modulation (PCM) packets are transmitted with high priority, while the least significant bits are transmitted with low priority.
An important step for PLC techniques arrived when Gundhuzan and Momtahan proposed in [4] an high performance packet loss concealement algorithm based on linear predictive coding (LPC). LPC is a technique widely used - also in this paper - that allows to estimate future samples by exploiting the statistical characteristics of the past ones. In LPC the signal is modelled as an autoregressive stochastic process composed of two contributions: a linear combination of past samples convoluted by an all poles filter and a scaled excitation signal.
Other statistical methods include Hidden Markov Chains and dictionary based schemes ([5] - [6]).
In the last years PLC methods mostly exploit predictors based on neural networks in order to reconstruct missing future packets from the context. We can group these methods into two categories: feed-forward models (fast models that can occasionally cause signal discontinuity) and models based on recurrent neural networks (slower but more accurate).
In the first family of methods we can find techniques of PLC based on feed-forward deep neural networks (DNNs) for digital speech transmission. For example, in [7] has been proposed DNNs with a multiple-layer deep architecture. Again, [8] proposed two methods of PLC combining DNNs with advanced signal processing techniques for phase correction. Among honorable mentions the literature proposes also packet loss concealment for networked music performance applications with convolutional layers ([9]).
Other methods of this type are based on non-autoregressive packet loss adversarial auto-encoder (PLAAE), to perform real-time PLC in the waveform domain ([10]).
A Basic PLC method based on Generative Adversarial Networks (GAN) is presented in [11]. GAN-based approaches have been shown to be able to efficiently generate audio waveforms [12] because a loss function doesn't need to be specified since it is internally learned by the network. A more advanced GAN based method is presented in [13], a temporal-spectral generative adversarial network based end-to-end (TF-GAN).
For what concern systems based on autoregressive neural networks with application to PLC we can find techniques based on adaptive recurrent neural network (RNN) in [14] and other sequential raw audio synthesis methods based on WaveRNN architecture ([15] - [16]).
Finally, LPCnet [17] improves neural speech synthesis through linear prediction using an hybrid neural PLC architecture where the missing speech is synthesized using a generative model conditioned by a predictive model.
LPCnet has been proposed during the last INTERSPEECH challenge 2022 [18]. The challenge consisted in presenting innovative PLC algorithms. The competitors trained and tested their models on a dataset released as part of the challenge, taken from a base public domain podcast dataset (LibriVox Community Podcasts).
In this report results will be evaluated on the same dataset in order to compare them with those of the participants to challenge. For this purpose we present an adaptation of the PARCnet model [19] - which was best suited for real-time music applications - applied to corrupted speech signals. PARCnet is composed of two parallel branches. One is an autoregressive linear predictor and the other is a

fully convolutional feed-forward neural network. Differently from other DNNs-based methods, it recasts the problem of predicting the whole packet to the prediction of the residual of the AR model. Since the neural module runs in parallel to the autoregressive one, it predicts only the residual while the rest of the signal is computed by the AR model that performs light-weight computations.

## 2. METHODS

As mentioned above, the main purpose of this report is to examine the effectiveness of [19], a PLC method proposed for networked music performance in the corrupted speech reconstruction domain. Hence, no major changes have been made to the original work. In the following sections we briefly describe the **PARCnet model**. The adopted method is composed of two cooperating modules: a Neural Network and a Linear AutoRegressive Predictor. Total execution time is hence dominated by the parallel AR branch.

### 2.1. Linear AutoRegressive Predictor

Assuming that a signal can be modelled as an autoregressive process, a sample $\hat{y}[n]$ can be predicted with the linear combination of the preceding samples, weighted by $\varphi_i$, plus an excitation signal:

$$\hat{y}[n] = \sum_{i=1}^{p} \varphi_i y[n-i] + \varepsilon[n] \tag{1}$$

Being the memory finite, the residual $\boldsymbol{\varepsilon}$ cannot be assumed as white noise. It cannot be known a priori and so the estimation of the prediction is strongly dependent of the previous samples ([4]) and rather slow: from here the idea to predict the residual with a feed-forward neural network from the past context, allowing a faster estimation of the missing samples.

As a result the missing packet estimation can be thought in two steps. The first is a purely autoregressive intermediate evaluation of the lost packet, while the second is the sum between the previous result $y[n]$ and the neural network inference $f_\vartheta(\mathbf{x})_{(n-k)}$:

$$\begin{cases} \hat{y}[n] = y_{AR}[n] + f_\vartheta(\mathbf{x})_{(n-k)} \\ y_{AR}[n] = \sum_{i=1}^{p} \varphi_i y[n-i] \end{cases} . \tag{2}$$

where $\mathbf{x}$ is the context vector of the past N samples, $n = k, ..., k+M$ with $M$ the length of the audio packet, $f_\vartheta : \mathbb{R}^N \rightarrow \mathbb{R}^M$ vector-valued neural mapping parameterized by $\vartheta$, $k$ the time index of the missing packet, $p$ the order of the linear predictor and $\varphi_i$ the weights associated to each sample of the linear prediction.
By taking into consideration a window comprehensive of a part of a valid packet preceding the loss and the reconstructed one, it is possible to reduce discontinuities between the two. In particular the amplitude discontinuities are minimized using cross-fading.
Phase discontinuities are modulated by weighting the contributions of the neural network and the autoregressive predictor by two envelopes $\mathbf{u}$ and $\mathbf{v}$ with values between 0 and 1:

$$\hat{\mathbf{y}} = \mathbf{u} \odot \boldsymbol{y_{AR}} + \mathbf{v} \odot f_\vartheta(\mathbf{x}), \tag{3}$$

$\mathbf{v}$ is a ramp fade-in vector used to suppress the audible discontinuities introduced by $f_\vartheta(\mathbf{x})$, while $\mathbf{u}$ is a ramp fade-out vector that removes the artifacts arising when the poles of the AR filter are near to unitary circle, since they could cause strong divergence.
Differently from the previous work, for the AR calculation of $\mathbf{y}$ the current algorithm uses explicit computation of the autocorrelation matrix $\mathbf{R}$ and of the autocorrelation vector $\mathbf{r}$ and then solves the Yule-Walker equations, i.e.:

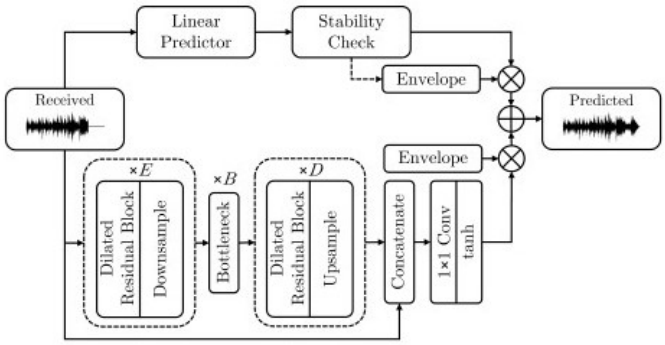



Figure 1: PARC-Net architecture, image from [19]

$$\mathbf{r} = \mathbf{R}\boldsymbol{\Phi}, \tag{4}$$

using the Levinson-Durbin recursion, with $\boldsymbol{\Phi}$ vector of $\varphi_i$

### 2.2. Neural Network architecture and Training

For the prediction of the residual we make use of a feed-forward neural network $F_\vartheta(\cdot)$. Considering the previous N valid samples to the packet loss at index k, we define an input vector $\tilde{\mathbf{x}}$ and append *M'* zeros to it. We obtain $\tilde{\mathbf{x}}$ and consequently the output vector $F_\vartheta(\tilde{\mathbf{x}})$ of the same dimensions of $\tilde{\boldsymbol{y}}_{\boldsymbol{AR}} \in \mathbb{R}^{N+M'}$, the corresponding vector of linearly predicted samples.
$\tilde{\boldsymbol{y}}_{\boldsymbol{AR}}$ and the output of $F_\vartheta(\tilde{\mathbf{x}})$ will be summed in order to predict the training objective $\tilde{\mathbf{y}} = [y[k-N], ..., y[k+M']]^T$, the windowed ground-truth waveform.

In such a way, the model learns from the ground-truth how to correctly reconstruct the excitation signal by means of the following minimization operation of the loss $\mathcal{L}$:

$$\min_\vartheta \mathcal{L}(\tilde{\mathbf{y}}, \tilde{\boldsymbol{y}}_{\boldsymbol{AR}} + F_\vartheta(\tilde{\mathbf{x}})) \tag{5}$$

Once the training is completed, the inference is purely in time domain and outputs a vector of *N+M'* time-domain samples.
For what concerns the Neural Network architecture, it is a fully convolutional model. The first stage is an encoder that reduces the representation of the input by means of a decreasing number of filters. To further compress the features map, a bottleneck is applied. Subsequently, a decoder reconstructs the modified representation through an increasing number of filters. The result of these stages is concatenated with the received input, convolved and fed as input to an Hyperbolic Tangent activation function. Simultaneously, the linear predictor of order $p$ reconstructs the missing packet in the other branch as shown in figure 1. Finally the output of the two branches are summed, to obtain the complete prediction of the missing packet.

## 3. EVALUATION

### 3.1. Dataset

The training has been done on a subset of the exact same dataset provided by the INTERSPEECH 2022 challenge on Audio Packet Loss Concealment by Microsoft, intended to stimulate research in the area of PLC. Accordingly, it has been possible to compare our results directly with partecipants' ones. The comparison will give an idea of the reliability of our model with respect to the other proposed state-of-the-art techniques.
The challenge provided a set of audio and meta-data files, subdivided into a train split, a validation split and blind test split.
The dataset has been created [18] by samples of real corrupted calls made by Microsoft Teams users. Segments of 10 seconds with at least one lost packet were randomly extracted from calls. Subsequently, a model has been trained in order to learn the audio losses distribution in an excerpt of 10 seconds. To create the dataset with packet losses, the model has been applied to randomly chosen segments from a podcast in english language (LibriVox Community Podcasts) with the effect of zeroing out some regions in the audio clips.
For every excerpt the validation and the blind test splits contain the clean sample file, the "lossy" audio sample with packet losses set to zero and a text file with loss metadata. The text files are supposed to indicate whether a packet has been lost. Each packet is represented with one line for each 20 milliseconds segment of audio: if line contains a "0" the packet has been lost, if contains "1" the packet is intact. Moreover, is provided a csv file containing metadata about the included clips. The complete training set includes a total of 23184 clips, while the validation and blind test sets include 966 clips each.

### 3.2. Metrics

To evaluate our results, two metrics has been used: PLCMOS (Packet Loss Concealment Mean Opinion Score) and WAcc (Word Accuracy). PLCMOS is the neural network presented by INTERSPEECH challenge and trained to estimate how a human listener would rate the quality of an audio excerpt. Thus, it can quickly evaluate the performance of PLC systems and was trained only on speech. The aforementioned is a convolutional-recurrent encoder. Two different versions can be exploited: a semi-intrusive model requiring a reference that does not need to be aligned with the lossy signal, and a non-intrusive one that does not need the reference audio. The algorithm needs in input signals resampled at 16kHz. PLCMOS gives a score between 1 and 5 as an estimate of human mean opinion scores and it has been proposed in order to substitute Crowd-Sourced Comparative Mean Opinion Score (CMOS), a metric used in INTERSPEECH challenge. However, it wasn't possible to exploit it because it needs an audience that gives a score to results' quality. It is worth to notice that PLCMOS and subjective CMOS show a Pearson correlation coefficient of 0.95 and a Spearman rank correlation coefficient of 0.93.
The WAcc is calculated using Microsoft Azure Cognitive Services of speech recognition by subtracting the returned Word Error Rate (WER) from 1. The score is between 0 and 1.
Word accuracy was calculated based on 966 clips of the blind set. 21 clips containing multi-lingual or non-speech audio were excluded from WER evaluation because the speech recognition service could not reliably produce transcriptions.

### 3.3. Training and Testing

As previously mentioned, the experiments have been run using a subset of INTERSPEECH 2022 dataset converted to mono and sampled at 16 kHz. For our purposes have been considered packets of length 20 ms (320 samples); it was done so because 20 ms is a value of buffer size just above the ones used by sound cards available on the market. In the training and validation phase are required a lossy signal to reconstruct and a reference one to compare it with. Even if the official dataset provides lossy and text files for validation purposes, in our case these are not used since they are created by the code itself during the training as explained below.
In the proposed algorithm, that exploits PyTorch functionalities, shorter signals are created from the challenge's clean dataset as follows: 10 slices of 51 packets were randomly extracted from each single 10 seconds long audio file. From those excerpts the actual lossy input data for the training are generated. They consist of 7 packets predicted with the aforementioned AR model (formula (4)), using a context of 5 packets (1600 samples, 100 ms), and one more set to zero. The final result is a not-normalized sequence 160 ms long. Clean data are created similarly, except for the last packet which is also predicted with the AR model. In this way 205.000 audio files (7 packets in the past + lost packet) has been extracted.
The training has been done over 100 epochs.

During training divergence problems have occurred. This is caused by the substantial differences between the nature of the original MAESTRO dataset, made of piano recordings, [19] and the one provided by INTERSPEECH Challenge [18] described in the previous paragraph. In particular, even just one packet incorrectly rebuilt by the AR process provokes the divergence of the whole training. For this reason a different AR model implementation - the one proposed in formula (4) - has been used at the expense of the original one, in order to better control the stability of the process. The main advantages of the new linear predictor are the followings: faster computation due to the usage of a C compiler and the possibility to control the parameters that ensure the reversibility of the autocorrelation matrix. Indeed, to compute the coefficients of the AR model filter is necessary to determine the autocorrelation matrix $\mathbf{R}$ and the autocorrelation vector $\mathbf{r}$ of the input signal and invert it. Despite this change, matrix diagonal loading continues to apply by summing a infinitesimal constant to the diagonal of the matrix. This allows to overcome non singularity problems and to improve the stability of the reconstruction process.

Subsequently, the obtained model was tested on the given blind set of 945 files (21 on 966 has been excluded because containing multi-lingual or non-speech audio). As seen in formula (3), the reconstruction of the lost packet consists in the sum of the linear prediction with the same training AR model (4) and the inference of the neural model, weighted with appropriate cross-fades at the beginning and at the end of each packet. The original reconstruction process was changed as a consequence of a different dataset. To deal with possible divergences of the autoregressive model, 400 samples are predicted, of which the last 80 are overlapped with the following packet and scaled by a fade-out. In the original version it was assumed that two subsequent packets couldn't be lost, which is possible in our case. This required that in such situations exactly one packet of 320 samples was predicted without any fade-out. Moreover, in the initial PARCNet it was assumed that there could

Table 1: results and table from 2022 challenge INTERSPEECH [18]. Teams marked as "tied": difference not significant ($p \geq 0.05$). Parameter count and structure given where known (TD: Time Domain, FD: Frequency Domain, FF: feed-forward, LPC: linear predictive coding)

| Rank | Team ID | Company | PLCMOS | WAcc | Parameters | Approach |
|---|---|---|---|---|---|---|
| – | – | Clean (reference) | 4.51 | 0.97 | – | – |
| 1 | 12 | Kuaishou | 4.28 | 0.88 | 2.36M | TD U-Net |
| 2 | 1 | Amazon | 3.74 | 0.88 | 5.9M | RNN + LPCNet |
| 3(tied) | 9 | Alibaba Inc. | 3.83 | 0.87 | – | TD Encoder-Decoder |
| 3(tied) | 14 | Oldenburg University | 3.98 | 0.86 | 4.97M | TD Sequence-to-One |
| 5 | 11 | Samsung | 3.28 | 0.86 | 36k | FD Conv. Net |
| 6 | 10 | UCAS | 3.48 | 0.83 | – | – |
| – | – | Zero-filling baseline | 2.90 | 0.86 | – | – |
| 7 | 6 | Shenzen University | 2.90 | 0.86 | 3.77M | – |
| – | – | Politecnico di Milano | 3.08 | 0.80 | 416k | LPC+FF Neural Net |

Table 2: results and table from 2022 challenge INTERSPEECH [18]: metrics for non-eligible models.

| Team ID | Company | PLCMOS | WAcc |
|---|---|---|---|
| 2 | GZHU-CUC | 4.406 | 0.854 |
| 13 | Augsburg University | 3.274 | 0.853 |
| 3 | VTCC | 3.371 | 0.852 |

be no losses in the first 7 packets or in the last one of the audio to be reconstructed. This is not true for the INTERSPEECH dataset. Thus, it was necessary to zero-pad inside the reconstruction in order to rebuild the packets in those positions.

### 3.4. Results and Discussion

The best and more recent checkpoint of the training (epoch 83) gives as result packet MSE of $-1.45332$ dB and total MSE on the output of the network of $-7.89$ dB.
After the testing phase we obtained as results a PLCMOS of 3.08 and a WAcc of 0.80. The PLC performance on a speech dataset of the proposed method has been compared with the several baselines proposed during the challenge of INTERSPEECH 2022 (table 1 and 2).
As mentioned above, the main goal of this work was to evaluate the reconstruction quality of PARCnet in another application domain, i.e. speech. As expected, the reconstruction's accuracy gave discrete results with respect to the baselines that were originally designed for speech. This is due to the substantial differences between a dataset of speech recordings and a dataset of music (instrument) recordings. Further studies and solutions on the PARCNet model are therefore needed to significantly increase its performance.
However, it should be noted that PARCnet has the second lowest number of trainable parameters among all the PLC models proposed by the challenge (416 k), i.e. less than a fifth of the first classified (2.36M) and less than a tenth of those of LPCnet (5.9M, runner-up). It is reasonable to think that it will outspeed all the other PLC techniques once a meticulous optimization is performed.

## 4. CONCLUSION

PARCnet is a new hybrid packet loss concealment method that combines a linear autoregressive model with a feed-forward neural network. It exploits the short-time statistical properties of signals to apply linear autoregression. In the parallel branch the neural predictor estimates the nonlinear residual term in a single step. In this way, performance is optimized because the neural part has the only task of reconstructing the excitation signal. The total execution time is dominated by the autoregressive branch and the speed of execution increases significantly. PARCNet was originally proposed for music signals for which it proved very effective, reaching state-of-the-art results and significantly outperforming recent deep learning based methods. However, our work shows that its performance is not on par with the baseline methods of INTERSPEECH challenge 2022 when it comes to speech signals. Simply moving from the application domain of real-time music to speech is not enough to get equally valid results. Further studies and improvements are needed to successfully adapt the model to the new application domain.

## 5. REFERENCES

[1] D. J. Goodman, G. B. Lockhart, O. J. Wasem, and L. W.-C. Wong, "Waveform substitution techniques for recovering missing speech segments in packet voice communications," *ICASSP '86. IEEE International Conference on Acoustics, Speech, and Signal Processing*, vol. 11, pp. 105–108, 1986.

[2] Y.-L. Chen and B. Chen, "Model-based multirate representation of speech signals and its application to recovery of missing speech packets," *Speech and Audio Processing, IEEE Transactions on*, vol. 5, pp. 220 – 231, 06 1997.

[3] J. Suzuki and M. Taka, "Missing packet recovery techniques for low-bit-rate coded speech," *Selected Areas in Communications, IEEE Journal on*, pp. 707 – 717, 07 1989.

[4] E. Gunduzhan and K. Momtahan, "Linear prediction based packet loss concealment algorithm for pcm coded speech,"

*Speech and Audio Processing, IEEE Transactions on*, vol. 9, pp. 778 – 785, 12 2001.

[5] C. Rodbro, M. Murthi, S. Andersen, and S. Jensen, "Hidden markov model-based packet loss concealment for voice over ip," *Audio, Speech, and Language Processing, IEEE Transactions on*, vol. 14, pp. 1609 – 1623, 10 2006.

[6] Y. Bahat, Y. Schechner, and M. Elad, "Self-content-based audio inpainting," *Signal Processing*, vol. 111, 06 2015.

[7] B.-K. Lee and J.-H. Chang, "Packet loss concealment based on deep neural networks for digital speech transmission," *IEEE/ACM Trans. Audio, Speech and Lang. Proc.*, vol. 24, no. 2, p. 378–387, feb 2016. [Online]. Available: https://doi.org/10.1109/TASLP.2015.2509780

[8] Q. Ji, C. Bao, and Z. Cui, "Packet loss concealment based on phase correction and deep neural network," *Applied Sciences*, vol. 12, no. 19, 2022. [Online]. Available: https://www.mdpi.com/2076-3417/12/19/9721

[9] P. Verma, A. Mezza, C. Chafe, and C. Rottondi, "A deep learning approach for low-latency packet loss concealment of audio signals in networked music performance applications," 07 2020.

[10] S. Pascual, J. Serrà, and J. Pons, "Adversarial auto-encoding for packet loss concealment," 07 2021.

[11] Y. Shi, N. Zheng, Y. Kang, and W. Rong, "Speech loss compensation by generative adversarial networks," in *2019 Asia-Pacific Signal and Information Processing Association Annual Summit and Conference (APSIPA ASC)*, 2019, pp. 347–351.

[12] J. Kong, J. Kim, and J. Bae, "Hifi-gan: Generative adversarial networks for efficient and high fidelity speech synthesis," 2020.

[13] J. Wang, Y. Guan, C. Zheng, R. Peng, and X. Li, "A temporal-spectral generative adversarial network based end-to-end packet loss concealment for wideband speech transmission," *The Journal of the Acoustical Society of America*, vol. 150, pp. 2577–2588, 10 2021.

[14] S. Pascual, J. Serrà, and J. Pons, "Adversarial auto-encoding for packet loss concealment," 07 2021.

[15] N. Kalchbrenner, E. Elsen, K. Simonyan, S. Noury, N. Casagrande, E. Lockhart, F. Stimberg, A. van den Oord, S. Dieleman, and K. Kavukcuoglu, "Efficient neural audio synthesis," 2018.

[16] F. Stimberg, A. Narest, A. Bazzica, L. Kolmodin, P. B. González, O. Sharonova, H. Lundin, and T. C. Walters, "Waveneteq — packet loss concealment with wavernn," *2020 54th Asilomar Conference on Signals, Systems, and Computers*, pp. 672–676, 2020.

[17] J.-M. Valin and J. Skoglund, "Lpcnet: Improving neural speech synthesis through linear prediction," 2019.

[18] L. Diener, S. Sootla, S. Branets, A. Saabas, R. Aichner, and R. Cutler, "INTERSPEECH 2022 Audio Deep Packet Loss Concealment Challenge," in *Proc. Interspeech 2022*, 2022, pp. 580–584.

[19] A. S. A. I. Mezza, M. Amerena, "Hybrid packet loss concealment for real-time networked music applications," *Unpublished*, 2023.